Yihao Xu

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Research Profile

- Computational design science researcher working on multimodal learning and predictive analytics for digital platforms, with additional work in computational measurement of organizational culture.
- First author of papers in the *International Conference on Information Systems* and *IEEE Sensors Journal*; the ICIS 2025 paper was recognized under the conference's Best Complete Paper Nominees / Best in Track Papers distinction.
- Experience designing multimodal and multi-view learning models, modeling temporal distribution shift, and measuring constructs with neural text embeddings; Python, PyTorch, and SQL.
- Teaching assistant across two fall terms; coordinated six study sessions involving 60 participants who are deaf or hard of hearing in an AR accessibility study.

Education

M.S. in Management Science and Engineering **Expected 2027**
School of Management, University of Science and Technology of China (USTC) *Hefei, China*
Advisor: Prof. Hefu Liu.

B.M. in Information Management and Information Systems **2024**
School of Management, Hefei University of Technology (HFUT) *Hefei, China*

Research Interests

Topics: Digital platforms and crowdfunding · Cultural and creative markets · Organizational culture · Accessible human–AI interaction
Methods: Multimodal representation learning · Multi-view deep clustering · Temporal generalization · Computational text measurement

Published Research

1. Subtly There, Deeply Felt: Modeling Cultural Values in Crowdfunding Success Prediction
Yihao Xu, Wenxiang Zhao, and Xiaoyu Yao.
46th International Conference on Information Systems (ICIS), Sharing Economy, Platforms, and Crowds Track, Nashville, TN, 2025.
ICIS 2025 Best Complete Paper Nominee / Best-in-Track Paper.

- **Situation:** Cultural crowdfunding projects carry cultural attributes that conventional pre-launch prediction models rarely represent explicitly.
- **Task:** Operationalize six Throsby cultural values from project metadata, text, and images for pre-launch success prediction.
- **Action:** Developed CV-PM, combining cultural-value mapping, deep representation refinement, and attention-based multimodal fusion.
- **Result:** On 14,539 Kickstarter projects, CV-PM achieved 90.86% ACC, 95.14% AUC, and 93.08% F1; AUC and F1 exceeded the strongest reported baselines by up to 3.65 and 3.18 percentage points, while revealing category-specific cultural-value contributions.

2. Multisensor Deep Clustering Networks With Joint Learning Strategy for Fault Diagnosis Considering Consensus-Specific Constraint
Yihao Xu, Mingfeng Lu, Gang Wang, and Fang Fang.
IEEE Sensors Journal, 25(11), 19921–19936, 2025. doi:10.1109/JSEN.2025.3560460.

- **Situation:** Unlabeled multisensor industrial data contain shared and sensor-specific signals, but conventional two-stage deep clustering underuses their consensus information.
- **Task:** Learn clustering-friendly consensus representations without discarding complementary information from individual sensors.
- **Action:** Developed MsDCN with a sensor-aware Commonality-net, sensor-specific Individuality-nets, and a joint consensus-specific clustering objective.
- **Result:** Across six gearbox and bearing datasets, MsDCN improved ACC over K-means by 19.83%–28.41% and significantly outperformed most comparison methods, showing the value of joint representation and clustering for unsupervised multisensor fault diagnosis.

Work in Progress

3. Multimodal Learning under Evolving Environments for Crowdfunding Outcome Prediction
Yihao Xu, Wenxiang Zhao, and Hefu Liu. *Planned submission to Information Systems Research in 2026.*

- **Research problem:** Predict crowdfunding outcomes before launch from campaign text and images as the relationship between project content and success changes over time.
- **Method:** Combines multimodal representation learning with temporal adaptation to integrate campaign content and adjust its predictive contribution across changing environments.
- **Results:** Evaluated on 54,306 Kickstarter campaigns across four categories against nine external methods. With validation-based model selection, AUROC exceeded the strongest benchmark in each category by 3.55–5.63 percentage points under a fixed chronological split and 20 training seeds per category.

4. From Anchor to Evidence: A Theory-Driven Model for Newly Launched Crowdfunding Success Prediction
Wenxiang Zhao, **Yihao Xu**, and Xiaoyu Yao. *Working Paper.*

- **Situation:** Newly launched projects lack post-launch feedback, while platforms expose cues sequentially through list and detail pages.
- **Task:** Model how backers form an initial anchor and revise it using intrinsic and relational evidence available at launch.
- **Action:** Developed a multimodal prediction model that connects initial impressions with project-level and relational evidence, drawing on anchoring-and-adjustment theory.
- **Result:** On 13,192 Technology and 19,744 Art projects, the model achieved 94.57% and 83.30% AUC, respectively, outperforming all reported baselines.

5. Scale-Anchored Neural Measurement of Firms' Competing Values
Research in progress; measurement stage completed.

- **Situation:** Survey-based culture measures offer limited longitudinal coverage; annual-report language differs from validated scale items.
- **Task:** Measure four Competing Values Framework orientations from corporate disclosures while retaining the OCAI scale's item structure.
- **Action:** Developed a scale-anchored text measurement approach and produced firm-year culture scores from 4.37 million MD&A segments (2001–2025).

6. AI Assistants and Customer Satisfaction in Digital Service Platforms
Early-stage Research Project.

- **Situation:** AI assistants increasingly mediate customer service, but satisfaction-relevant interaction patterns remain insufficiently understood.
- **Task:** Examine how conversational behavior in Meituan customer–AI assistant dialogues relates to user satisfaction.
- **Action:** Organize dialogue records around interaction behaviors and satisfaction outcomes for subsequent computational analysis.

Conference Presentation

Yihao Xu, Wenxiang Zhao, and Xiaoyu Yao. *Subtly There, Deeply Felt: Modeling Cultural Values in Crowdfunding Success Prediction.* 46th International Conference on Information Systems (ICIS), Sharing Economy, Platforms, and Crowds Track, Nashville, TN, Dec. 2025.

Research and Teaching Experience

Research Assistant, University of Science and Technology of China *Jun. 2026–Present*
Supported an AR smart-glasses study on communication access and social inclusion for people who are deaf or hard of hearing; coordinated materials, participant arrangements, device setup, and six study sessions involving 60 participants.

Teaching Assistant, University of Science and Technology of China *Fall 2025, 2026*
Information Technology and Organizational Strategy 42 students; 40 hours
Instructor: Prof. Juan Qin; coordinated assignments, group activities, and feedback.

Awards and Scholarships

Awards

- Outstanding Graduate of Anhui Province (top 3% students) 2024
- Outstanding Student Award (top 10% students) 2023

Scholarships

- China National Scholarship 2025
- First-Class Graduate Scholarship, USTC 2025, 2026
- Shishengqing Scholarship, School of Management, HFUT 2024
- Academic Progress Scholarship, School of Management, HFUT 2022
- First-Class Scholarship, HFUT (top 3% students) 2021

Internships

Research Intern *Jun.–Aug. 2026*
USTC Yangtze River Delta Science and Technology Strategic Frontier Research Center
Conducted desk research, case analysis, and comparative studies of firms' early development and commercialization in commercial spaceflight, AI, and semiconductors.

Technical Service Intern, CEO Office, Volkswagen (Anhui) Automotive Co., Ltd. *Apr.–Jul. 2024*
Hefei, China. Built and maintained an automotive after-sales database for recurring service reporting and operational analysis.

Computational Skills and Languages

Computational Methods: Multimodal learning · Multi-view deep clustering · Attention mechanisms · Temporal generalization · Sentence embeddings and semantic retrieval
Programming: Python · PyTorch · SQL · Stata
Languages: Chinese (native) · English (IELTS Academic 7.5)